Algebra SEP Worksheet 1 7/18/23

Linear Algebra

1. (August 2019 Problem 4) This problem involves 4×4 matrices with entries in $\mathbb{R}$. We will say that a matrix M is an E**-matrix** if M has 1's along the diagonal and a single nonzero entry off the diagonal. For instance:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 31 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

is an E-matrix. Express the matrix A below as a product of E-matrices.

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}$$

1. Solution This is a straight up row reducing problem. Let $E_{i,j}$ be the matrix with 1s on the diagonal, a 1 in position (i, j), and 0s elsewhere. Then

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix} = E_{2,1}E_{3,1}E_{4,1}E_{1,2}E_{1,3}E_{1,4}$$

2.

(a) Give an example of a finite dimensional vector space V over $\mathbb{C}$, and a nonzero linear transformation $A\colon V \to V$ whose only eigenvalue is 0.

(b) Prove that the only eigenvalue of $A\colon V \to V$ is 0 if and only if A is nilpotent. (V is still f.d. over $\mathbb{C}$.)

2. Solution

(a) Let $V = \mathbb{R}^2$, and A be the linear transformation corresponding to the matrix

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

(b) If A is nilpotent, then it satisfies the polynomial $x^k = 0$ for some k. But then the minimal polynomial of A divides x^k, and so its only eigenvalue is 0.

On the other hand, suppose the only eigenvalue of A is 0. Then the characteristic polynomial of A is a degree n polynomial whose only root is 0, which means it must be x^n. By the Cayley-Hamilton Theorem, A satisfies its own characteristic polynomial, so $A^n = 0$.

3. Give an example of a nonzero finite-dimensional vector space V over $\mathbb{C}$ and an invertible linear transformation $A\colon V \to V$ such that A is not diagonalizable.

3. Solution The classic example is the transformation:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

One way to think about Jordan normal form is that it shows that shearing is *exactly* the obstruction to being diagonalizable.

4. (August 2020 Problem 2) Let M be an $n \times n$ matrix with real entries. Let M^T be its transpose. Each of these two matrices defines a linear transformation $\mathbb{R}^n \to \mathbb{R}^n$; let A denote the transformation corresponding to M, and let B denote that corresponding to M^T.

(a) Prove that any vector in $\ker(A)$ is orthogonal to any vector in $\mathrm{im}(B)$ with respect to the standard inner product (dot product).

(b) Prove that $\mathbb{R}^n$ is the direct sum of $\ker(A)$ and $\mathrm{im}(B)$.

4. Solution

(a) Let $v \in \ker(A)$ and $u' \in \mathrm{im}(B)$. Then there is some $u \in \mathbb{R}^n$ so that $u' = M^T u$. Then we have $v \cdot u' = v \cdot (M^T u) = Mv \cdot u = 0 \cdot u = 0$.

(b) By the rank-nullity theorem,

$$n = \mathrm{rank}(M) + \mathrm{null}(M) = \mathrm{rank}(M^T) + \mathrm{null}(M) = \dim \mathrm{im}(B) + \dim \ker(A).$$

Also, since any vector in $\ker(A)$ is orthogonal to any vector in $\mathrm{im}(B)$, $\ker(A) \cap \mathrm{im}(B) = \{0\}$. Thus, $\mathbb{R}^n = \ker(A) + \mathrm{im}(B)$.

5. (August 2014 Problem 3 (a)-(c))

(a) Let V be a finite-dimensional vector space over $\mathbb{C}$, and let $T\colon V \to V$ be a linear transformation. Show that T has an eigenvector.

(b) Give an example of a nonzero finite-dimensional vector space V over $\mathbb{R}$ and a linear transformation $T\colon V \to V$ such that T does not have an eigenvector.

(c) Does a linear transformation of an infinite-dimensional vector space over $\mathbb{C}$ have to have an eigenvector? Either prove this is the case, or give an example of a linear transformation of an infinite-dimensional vector space with no eigenvector.

5. Solution

(a) Let $\chi(x)$ be the characteristic polynomial of T. Then by the fundamental theorem of algebra, χ has a root λ. This means that $\det(T - \lambda \mathrm{Id}) = 0$, so $T - \lambda \mathrm{Id}$ has nontrivial kernel. For any $v \in \ker(T - \lambda \mathrm{Id})$, we have $Tv = \lambda v$, so v is an eigenvector for T.

(b) The classic example is a rotation by $\frac{\pi}{2}$ radians:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

(c) This does not have to be the case. Consider the vector space V whose elements are sequences of complex numbers with only finitely many nonzero terms. So a vector looks like $v = (z_1, z_2, \ldots, z_t, 0, 0, 0, \ldots)$, where t might be different for different vectors v, but is always finite. Let T be the "right shift" operator, so $Tv = (0, z_1, z_2, \ldots, z_t, 0, \ldots)$. If we try setting up $Tv = \lambda v$, then looking at the first entry tells us the only possible value of λ is 0, but T has no kernel so it can't have any eigenvectors.

6. Let V and W be finite dimensional vector spaces over the same field (for the sake of argument, let's say over $\mathbb{R}$). Let $T\colon V \to V$ and $U\colon W \to W$ be linear maps. Consider the linear map $T \oplus U\colon V \oplus W \to V \oplus W$ which takes (v, w) to (Tv, Uw).

(a) Let $\mathcal{B}_1 = \{v_1, \ldots, v_n\}$ be a basis for V, and $\mathcal{B}_2 = \{w_1, \ldots, w_m\}$ be a basis for W. If A is the matrix of T with respect to $\mathcal{B}_1$, and B is the matrix of U with respect to $\mathcal{B}_2$, what is the matrix of $T \oplus U$ with respect to the basis $\mathcal{B} = \{(v_1, 0), \ldots, (v_n, 0), (0, w_1), \ldots, (0, w_n)\}$?

(b) What are the eigenvalues of $T \oplus U$ in terms of the eigenvalues of T and the eigenvalues of U?

(c) What is the trace of $T \oplus U$ in terms of the trace of T and the trace of U?

(d) Same question for the determinant of $T \oplus U$.

(e) Same question for the characteristic polynomial $\chi_{T\oplus U}$ of $T \oplus U$.

(f) Let $\mu_T(x)$ be the minimal polynomial of T, and $\mu_U(x)$ be the minimal polynomial of U. Prove that the minimal polynomial of $T \oplus U$ is $\operatorname{lcm}(\mu_T, \mu_U)$.

6. Solution

(a) The matrix of $T \oplus U$ with respect to $\mathcal{B}$ is the block diagonal matrix

$$\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}.$$

(b) If $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of T, and $\lambda_{n+1}, \ldots, \lambda_{n+m}$ are the eigenvalues of U, then $\lambda_1, \ldots, \lambda_{n+m}$ are the eigenvalues of $T \oplus U$.

(c) $\mathrm{Tr}(T \oplus U) = \mathrm{Tr}(T) + \mathrm{Tr}(U)$

(d) $\det(T \oplus U) = \det(T)\det(U)$

(e) $\chi_{T\oplus U}(x) = \chi_T(x)\chi_U(x)$

(f) Let $f = \operatorname{lcm}(\mu_T, \mu_U)$. Since both $\mu_T \mid f$ and $\mu_U \mid f$, we certainly have that $f(T \oplus U) = 0$. Now, let g be any other polynomial such that $g(T \oplus U) = 0$, and we will show that $f \mid g$. Since $g(T \oplus U) = 0$, then for any $v \in V$, $g(T \oplus U)(v, 0) = (g(T)v, g(U)0) = (0, 0)$, so $g(T) = 0$, and thus $\mu_T \mid g$. Similarly, $\mu_U \mid g$, and hence $f \mid g$.

7. (January 1983 Problem 1) Suppose V is a finite dimensional vector space over $\mathbb{C}$, and $T\colon V \to V$ is a linear operator. Suppose the subspace of V spanned by all eigenvectors of T is 1-dimensional. Prove that the minimal polynomial of T is $(x-\alpha)^n$, where $n = \dim V$ and α is some complex number.

7. Solution Not using Jordan normal form: the characteristic polynomial of T is a degree n polynomial whose only root is α, so it must be $(x-\alpha)^n$. By the Cayley-Hamilton theorem, T satisfies its minimal polynomial, so the minimal polynomial of T divides $(x-\alpha)^n$. We just need to check that $(T-\alpha\mathrm{Id})^k \neq 0$ for any $k < n$. Notice that $\dim\ker(T-\alpha\mathrm{Id}) = 1$, and for any linear transformations A, B, $\dim\ker(AB) \leq \dim\ker(A) + \dim\ker(B)$. Thus, $\dim\ker((T-\alpha\mathrm{Id})^k) \leq k < n$, so $(T-\alpha\mathrm{Id})^k \neq 0$.

Using Jordan normal form: if we take the Jordan normal form of T, all the Jordan blocks have α down the diagonal. However, if there is more than one block, then T will have two linearly independent eigenvectors, so actually the Jordan form of T is

$$\begin{pmatrix} \alpha & 1 & 0 & \dots & 0 \\ 0 & \alpha & 1 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & \alpha & 1 \\ 0 & \dots & \dots & 0 & \alpha \end{pmatrix}.$$

From here, one can compute that $(T-\alpha\mathrm{Id})^n = 0$, but $(T-\alpha\mathrm{Id})^k \neq 0$ for any $k < n$ (Indeed, for any upper triangular $n \times n$ matrix W with 0 on the diagonal, $W^n = 0$).

8.

(a) Give an example of two linear transformations with the same characteristic polynomial but different minimal polynomials.

(b) Give an example of two linear transformations with the same minimal polynomials which are not similar.

8. Solution

(a) Consider

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

(b) Here's where 6 (f) comes in handy. Consider

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

These matrices are not similar as they are both in Jordan Canonical form, but their minimal polynomials are both $(x-1)^2$.

9. (August 2014 Problem 3 (d)) Suppose that T and U are two linear transformations on a finite-dimensional vector space V over $\mathbb{C}$ which satisfy $TU = UT$. Prove that there is some $v \in V$ which is an eigenvector for both T and U.

9. Solution Let λ be an eigenvalue for T, and let $W \subset V$ be the subspace consisting of all eigenvectors for λ (and the zero vector). (We know we can find such things by 5 (a).) Here's the standard trick: for any $w \in W$, we have that $TUw = UTw = \lambda Uw$, so $Uw \in W$. Thus, we can think about U as a linear transformation on the smaller vector space W, and by 5 (a), U must have an eigenvector w'. But then w' is an eigenvector for both U and T, as desired.